Solar ESS Co. Ltd.
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ELECTRICIAN JOB SHEET · PROJECT BASED · QUOTE 2 OPTIONS

Battery backup for the water pump and lodge loads: manual ATS (Option A) or MTS for the pump (Option B)

Harmony@Huailan Ecolodge, Chiang Mai · Job no. JOB-HHL-PUMP-01 · Issued by Solar ESS Working Group · 2 Nov 2026

How it is wired today

- PEA grid → meter → **inverter GRID port**, and from the same point to the **MDB → all loads** (water pump, water heater and the other circuits).
- The inverter **BACKUP (LOAD) port** carries only 30–130 W. In a blackout the pump and all loads go off.
- The inverter can supply **10 kW total / 5 kW per phase** on backup (2× surge for 10 s), but the water heater alone draws 11–12 kW.

Option A: ATS at the MDB, kept in manual – back up the whole MDB

- PEA grid (after the inverter GRID connection) → **ATS source A**.
- Inverter BACKUP port → new MCB 4P 32 A → **ATS source B**.
- ATS output → MDB main breaker → all loads, including the pump.
- **Keep the ATS in MANUAL mode**: the total load can be higher than the inverter can supply. In a blackout, switch off the water heater and other big loads first, then turn the ATS to B.

Option B: 63 A MTS (manual I–0–II) for the water pump only

- Inverter BACKUP port → new MCB 2P 20 A → **MTS position II**.
- Existing pump MCB in the MDB (grid) → **MTS position I**.
- MTS output → RCBO 30 mA → water pump.
- In a blackout, turn the MTS to II to run the pump from the battery; when the grid is back, turn it back to I. Reference switch: Schneider GoPact MTSE 63 A open transition (I–0–II) or equal.

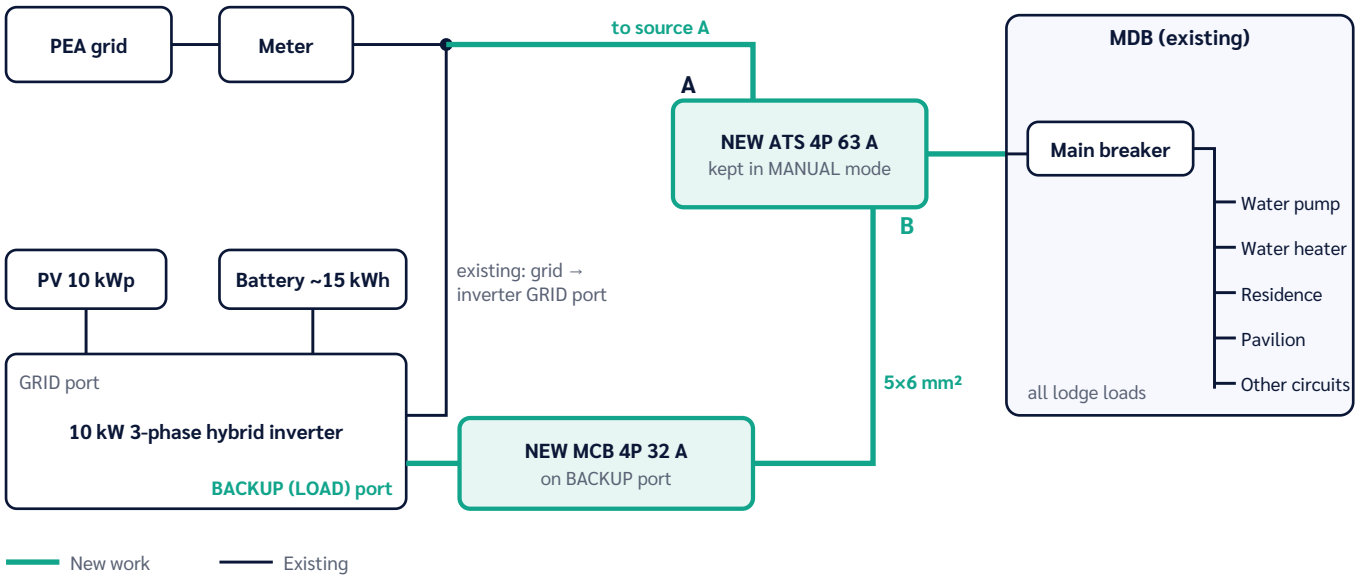
Rules that must never be broken: never connect the BACKUP port in parallel with the grid. Every switch must be break-before-make with a mechanical interlock and must switch the neutral too. Never connect the inverter GRID port to the output side of the ATS/MTS.

Existing system on site

Item	Details
Solar panels	10 kWp
Inverter	10 kW three-phase low-voltage hybrid · BACKUP port max 10 kW total / 5 kW per phase
Battery	Lithium 48 V 300 Ah (~15 kWh) · set to discharge down to 25%
MDB	Steel MDB with volt/ammeters; feeders: Residence, Pavilion, Guest House, Outside Toilet and spares
Big loads	Two-phase water heater ~11–12 kW (a few minutes a day) and the water pump (confirm rating on site)

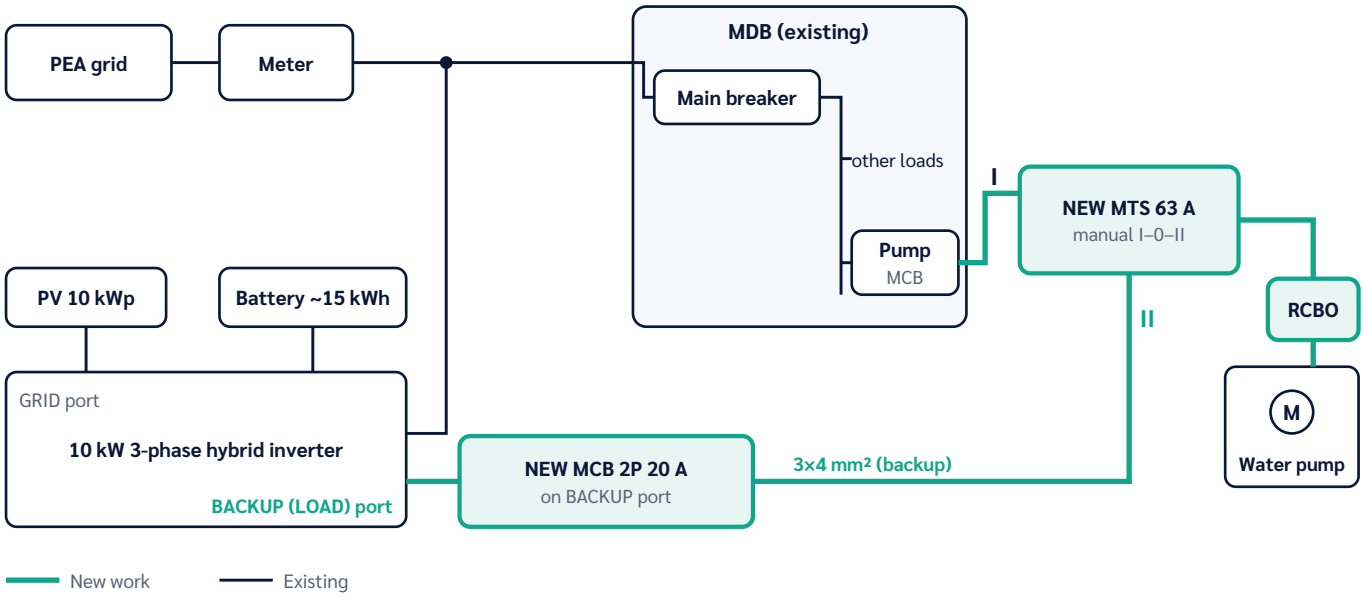
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Single-line diagram – Option A



The 4P ATS must be rated at least as high as the MDB main breaker (63 A minimum). Cables to source A and from the ATS to the MDB: same size as the existing main cable. Phase sequence on sources A and B must match.

Single-line diagram – Option B



Shown for a single-phase pump. For a three-phase pump use all 4 poles of the MTS, a 4P MCB and 5-core cable.

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Scope of work – Option A

1. **Survey:** MDB main breaker and main cable size, where the inverter GRID port is connected, inverter CT position, ATS location and cable runs. Send to Solar ESS before buying materials.
2. **Isolate (LOTO):** grid main, inverter GRID and BACKUP AC breakers, inverter off, battery breaker and PV DC isolators. Prove dead.
3. **ATS:** install a 4P ATS (≥ 63 A and not below the MDB main breaker) in an IP54 enclosure beside the MDB. **Set it to MANUAL** and disable automatic transfer.
4. **Source A:** move the MDB incoming main cable (downstream of the inverter GRID connection) to ATS source A. The inverter GRID port must stay connected directly to the grid as today.
5. **Source B:** install a new MCB 4P 32 A on the BACKUP port and run THW 5×6 mm² (L1 L2 L3 N PE) in 1" IMC/EMT to ATS source B.
6. **ATS output:** cable, same size as the existing main cable, to the MDB main breaker.
7. **Inverter CTs:** must stay on the grid cable upstream of the ATS (source A side). Check CT direction afterwards.
8. **Neutral:** the BACKUP-port neutral passes through the ATS only; never join it directly to the MDB neutral bar. PE may be common.
9. **Operating label at the ATS:** "Blackout: 1) switch off the water heater and big loads 2) turn ATS to B · Grid back: 1) turn ATS to A 2) heater on" and "Backup max 10 kW / 5 kW per phase".
10. **Test and hand over** per the test table, and show the lodge owner how to operate it.

Scope of work – Option B

1. **Survey:** pump nameplate (kW, V, 1- or 3-phase, starting method), starting current, pump circuit in the MDB and cable runs.
2. **Isolate (LOTO)** as in Option A.
3. **MTS:** install a 63 A MTS (I–0–II, open transition, e.g. Schneider GoPact MTSE or equal) in an IP54 enclosure at the MDB or near the pump.
4. **Position I:** move the outgoing cable of the existing pump MCB in the MDB to MTS position I.
5. **Position 2:** new MCB 2P 20 A on the BACKUP port (least-loaded phase, L1 recommended); THW 3×4 mm² in ¾" conduit to MTS position II (6 mm² if over 30 m).
6. **Output:** MTS → RCBO 1P+N 20 A 30 mA → water pump; earth the pump.
7. **Label:** "Blackout: turn to II (battery) · Grid back: turn to I (grid)".
8. **Test and hand over** per the test table, and show the lodge owner how to operate it.

> **Danger:** the inverter BACKUP port stays live from the battery and the solar panels even when the grid is off. Always switch off the inverter, the battery and the DC isolators first, lock out and tag out (LOTO), and prove dead. Tell the lodge owner in advance: Option A needs the whole lodge switched off while the main cable is moved.

Materials (minimum specification)

Premium materials only. Give the brand and size of every item in your quote.

Option	Item	Minimum specification	Qty
A	ATS	4P ≥ 63 A (not below the MDB main), 400 VAC, mechanical interlock, break-before-make, manual mode	1
A	ATS enclosure	Steel IP54 with DIN rail and terminals	1
A	MCB on BACKUP port	4P 32 A curve C 6 kA	1
A	Cable	THW 6 mm² (5 cores) BACKUP port → ATS; main-size cable for source A and ATS output	As measured
A	Conduit	1" IMC/EMT and main-cable size, with fittings, saddles and pull boxes	As measured
B	MTS	63 A 415 V, I–0–II open transition, IEC 60947-6-1, AC-23A (ref. Schneider GoPact MTSE 63 A or equal); 4P, or 2 poles used for a 1-phase pump	1
B	MTS enclosure	IP54 with DIN rail and terminals	1
B	MCB on BACKUP port	2P 20 A curve C 6 kA (to suit the actual pump)	1
B	RCBO, pump side	1P+N 20 A 30 mA Type A	1
B	Cable and conduit	THW 4 mm² (black, blue, green/yellow) in ¾" IMC/EMT; waterproof flexible connection at the pump	As measured
A/B	Consumables	Lugs, ferrules, cable markers, operating and warning labels, cable ties, anchors	1 set

Tests (record every reading)

Option	Test	Pass when
A/B	Insulation of every new cable at 500 VDC	≥ 1 MΩ
A/B	Polarity and earth continuity	Correct
A	Phase sequence of source A and source B is the same (essential for 3-phase motors/pumps)	Same
A	ATS on A: all loads on the grid, inverter normal, CTs read in the right direction	Normal
A	Simulated blackout: open the grid main, heater off, turn ATS to B; run pump and lights 15 min; record power per phase	≤ 5 kW/phase, no trip
A	Grid back: turn ATS to A; inverter resynchronises with the grid	Normal
B	MTS on I: pump runs on the grid as before	Pump normal
B	Simulated blackout: open the grid main, turn MTS to II; pump runs on battery 10 min, restart 3 times, record starting current	No trip
B	RCBO test button	Trips at once

To confirm on site

- MDB main breaker rating and main cable size (sets the Option A ATS size).
- Water pump: kW, 1- or 3-phase, starting current (if above 2.2 kW, tell Solar ESS first).
- Inverter CT position and the existing terminal block (labels RM1, SM1, TM1, RKM, SKM, TKM, LLM, N): identify every conductor before touching it.
- Cable runs and the ATS/MTS mounting position.

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How to quote

- Quote **Option A and Option B separately**.
- **Two prices for each option:** 1) Labour cost only; 2) Labour & materials (list every material by brand and size).
- Pay guide for labour: **Option A ฿6,000** (2 people, ~2 days) · **Option B ฿3,000** (2 people, ~1 day); materials and travel not included.
- Attach your electrician certificates and photos of similar work. Do not change any inverter settings; Solar ESS sets the inverter remotely.
- Send your quote with the form on www.solaress.org/job.html.

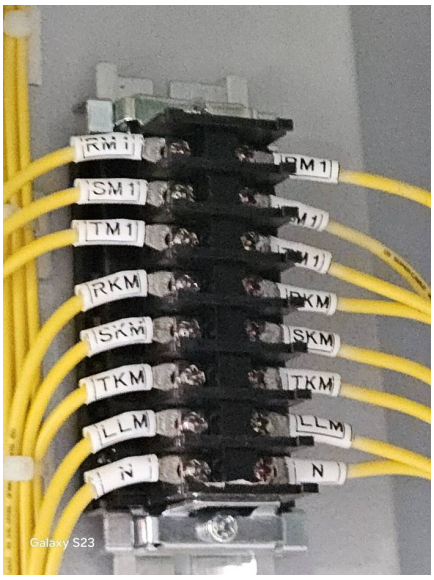
Site photos



Existing MDB (the ATS/MTS goes here)



Inverter room with AC and PV boxes



Existing terminal block (identify before work)

Qualifications and certificates of the Solar ESS team are furnished upon request or upon completion of contract.

Solar ESS Working Group